```
import pandas as pd
import matplotlib.pyplot as plt

data= pd.read_excel(r"C:\Fam\42\CSE 4202\lab1.xlsx")
print("The dataset is loaded, it has %d rows and %d columns." %
data.shape)
```

```
The dataset is loaded, it has 15 rows and 2 columns.
```

```
data.head()
```

```
   Height (FT)  Weight (KG)
0          5.3           60
1          5.6           63
2          5.4           62
3          5.1           65
4          5.1           60
```

```
x= data['Height (FT)'].values
y= data['Weight (KG)'].values

print(x)
print(y)
```

```
[5.3 5.6 5.4 5.1 5.1 5.6 5.8 4.8 4.9 4.5 5.4 5.7 5.1 5.9 5.2]
[60 63 62 65 60 70 75 72 74 73 78 71 70 66 69]
```

```
n= data.shape[0]
print(n)
```

```
15
```

```
sumx= x.sum()
sumy= y.sum()
sumx2= (x**2).sum()
sumxy=(x*y).sum()

print(sumx, sumy, sumx2, sumxy)
```

```
79.4 1028 422.44000000000005 5437.9
```

```
m= (n*sumxy - (sumx * sumy)) / (( n* sumx2) - (sumx)**2)
print(m)
```

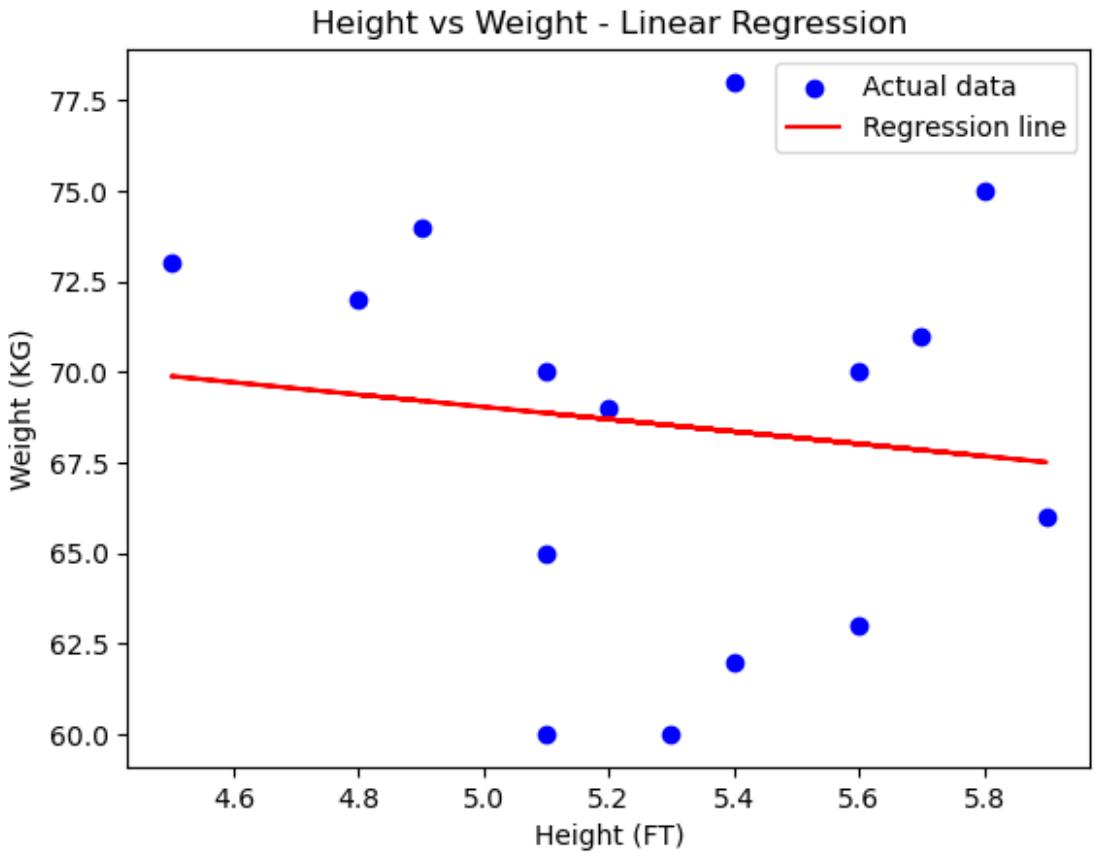
```
-1.6966501240698515
```

```
c= (sumy - (m*sumx)) /n
print(c)
```

```
77.51426799007642
```

```
plt.scatter(x,y, color='blue', label= 'Actual data')
plt.plot(x, m*x+c, color='red', label= 'Regression line')
```

```
plt.xlabel('Height (FT)')
plt.ylabel('Weight (KG)')
plt.title('Height vs Weight - Linear Regression')
plt.legend()
plt.show()
```




```
ypred= m*x+c
error=y - ypred
mse=(error**2).sum()/n
rmse=mse**0.5

print(mse, rmse)
```

```
29.036414392059566 5.3885447378730715
```

```
ss_res=((y-ypred)**2).sum()
ss_tol=((y-y.mean())**2).sum()
r2=1- (ss_res/ss_tol)

print(r2)
```

```
0.014006453635164062
```

```python
# Using builtin function/model.

import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Reshape x if it is a 1D array
X = x.reshape(-1, 1)

# Train the model
model = LinearRegression()
model.fit(X, y)

# Predict values
y_pred = model.predict(X)

# R² Score
r2 = model.score(X, y)
print("R² Score:", r2)

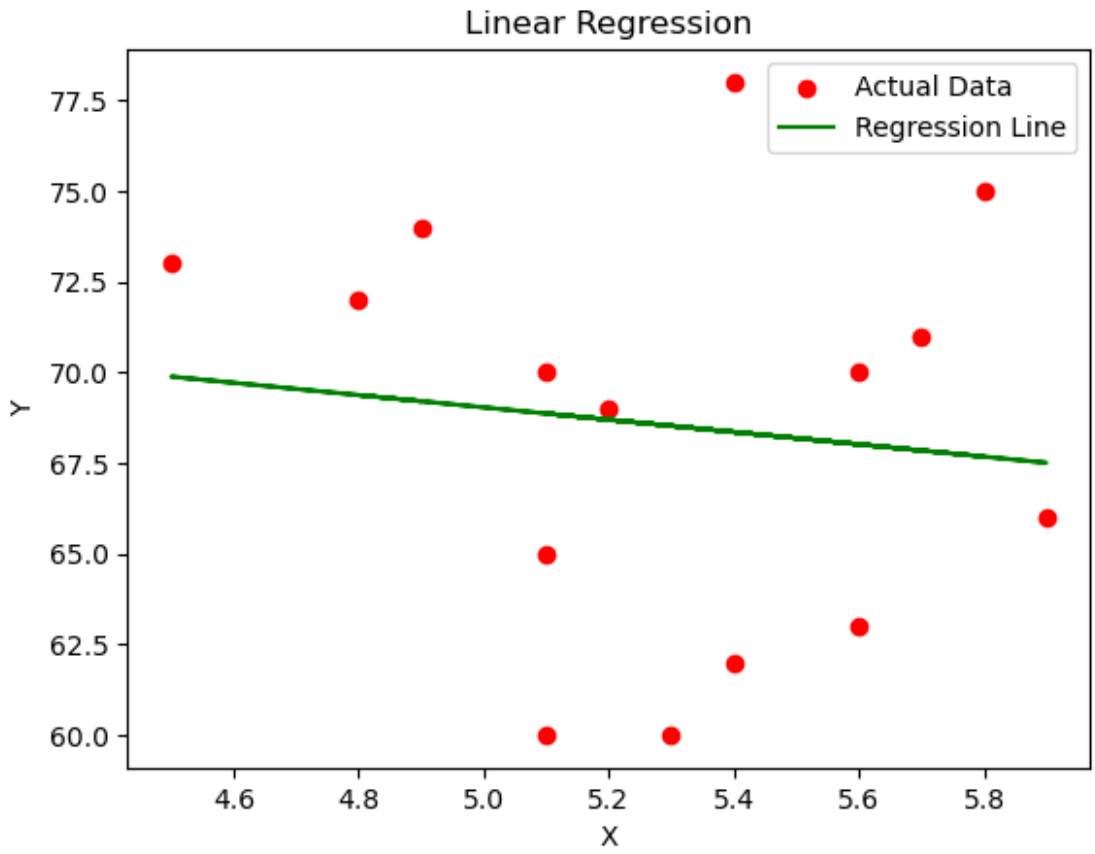
# Plot actual data
plt.scatter(X, y, label="Actual Data", color='Red')

# Plot regression line
plt.plot(X, y_pred, label="Regression Line", color='Green')

# Labels and title
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear Regression")
plt.legend()

# Show plot
plt.show()
```

```
R² Score: 0.014006453635164062
```


Linear Regression
Actual Data
Regression Line
77.5
75.0
72.5
70.0
67.5
65.0
62.5
60.0
Y
4.6
4.8
5.0
5.2
5.4
5.6
5.8
X